

# ENCN423/ENCN627 - Sustainable Energy Systems

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## Module 9: Transport

### Summary

Transport is a major contributor to global emissions, with solutions requiring both technological and behavioural shifts. Active and public transport, vehicle electrification, and smarter urban planning can significantly reduce energy use. Efficiency varies greatly by mode, with cycling and electric rail outperforming planes and private cars. Losses (acceleration, air drag) and vehicle size matter. For freight and aviation, electricity-based fuels are emerging, though challenges remain in cost and practicality. Policy and infrastructure choices will shape the path toward low-emission transport systems.

### Objectives

After this module, you should:

1. Understand the role of sustainable transport in defossilizing global and regional (e.g., New Zealand) energy systems.
2. Compare key strategies and technologies for transport defossilization, including electrification, biofuels, and hydrogen.
3. Assess the current global status of the transportation sector, including electric vehicle adoption and infrastructure.
4. Explain the fundamentals of electric vehicle architectures.
5. Evaluate operational challenges of transitioning different transport modes.
6. Develop and communicate evidence-based arguments for or against different defossilization pathways (e.g., the scalability of biofuels vs. the potential of synthetic fuels).

### Literature

#### Recommended:

- MacKay. Without the hot air. 2009. Chapters 3, 20, A (Cars, Better transport, Cars II). *Excellent overview on the modes of transport.*

- Sterner and Stadler - Chapter transport

**Do you want to learn more?:**

- IEA (2025), Global EV Outlook 2025, <https://www.iea.org/reports/global-ev-outlook-2025>
- CarbonBrief (2025), Factcheck: 21 misleading myths about electric vehicles, <https://interactive.carbonbrief.org/factcheck/electric-vehicles/index.html>

## Key Points

**Motivation:** Transportation accounts for 16.9% of global greenhouse gas emissions, with similarly significant contributions in New Zealand (and much higher in terms of CO<sub>2</sub> emissions). We can examine transportation sustainability through the energy trilemma framework:

- **Environmental**, e.g. emissions reduction (CO<sub>2</sub>, air/noise pollution), resource efficiency, and land use optimization.
- **Social**, e.g. road safety, public health (active mobility), affordability, and community connectivity.
- **Economic**, e.g. cost-effective mobility systems and local economic development.

The focus will center on energy efficiency and defossilization pathways across transport modes, while acknowledging these interconnected sustainability dimensions.

**Strategies for sustainable transportation:** We can reduce the environmental impact of transport through **technology, behaviour, and system design**. A key classroom discussion focused on estimating personal transport demand and energy use, highlighting challenges. In public transport we mentioned underutilization, poor maintenance, slow connections, and crowding.

We examined energy use versus speed across transport modes, noting high efficiency in cycling and electric trains, and poor efficiency in modes like planes, and hydrogen cars. The main energy drains? Acceleration (especially in short trips with frequent stops) and air resistance (which dominates in long trips). To cut energy waste, we can reduce vehicle weight, increase travel distance between stops, move slower, and use regenerative braking. According to MacKay's principles of sustainable transport, we should:

- Cut down vehicle size and weight per person
- Improve drivetrain efficiency
- Stick to steady speeds
- Travel at lower speeds
- Reduce unnecessary travel
- Ride-share to increase vehicle occupancy

But there's more! To really make a dent, we need electric vehicles, smarter charging solutions, promote active transport (walking and cycling), boost public transport, and reduce overall demand through remote working and better urban planning. Can you think of any others?

**Update on transport status:** Global transport is undergoing a major transformation, driven by electrification, declining fossil fuel use, and evolving technology trends.

- **Common units:**

- Passenger-kilometre (p-km): the transport of one passenger over one kilometre; used to quantify personal mobility.
- Tonne-kilometre (t-km): the transport of one tonne of goods over one kilometre; used in freight and cargo metrics.

- **Fuel use projections:**

- Fossil fuel use in both passenger and freight transport is projected to decline significantly.
- Electromobility is expected to become the dominant energy source, with electricity-based fuels playing a smaller, specific role.

- **Electric Vehicle (EV) growth trends:** EV sales are increasing rapidly worldwide, with the strongest growth in China, the Nordics, Ethiopia, Nepal and a few others in Southeast Asia and Central and South America. Following an S-curve, EV adoption is accelerating—driven by advances in batteries.

- **Affordability and emissions:** EVs are approaching sticker-price parity with internal combustion engine (ICE) vehicles. Many government subsidies aim to bridge the remaining cost gap. With clean electricity, EVs could achieve near-zero lifecycle emissions. Hydrogen fuel cell cars have seen minimal uptake globally, even in “hydrogen-friendly” countries like Japan.

**Car architecture:** vehicles come in various architectures depending on how energy is stored and delivered to the wheels.

- **Conventional vehicles** rely entirely on internal combustion engines (ICE) powered by petrol or diesel.
- **Hybrid vehicles:** combine a combustion engine with an electric motor.
  - Parallel hybrids use either power source to drive the wheels.
  - Plug-in hybrids allow battery charging from the grid and can operate in electric-only mode.
  - Series hybrids use the ICE to generate electricity for an electric motor, which drives the wheels.
- **Battery Electric Vehicles (BEVs)** use only a battery and electric motor, with no fuel tank or combustion engine.

- **Fuel Cell Electric Vehicles (FCEVs):** generate electricity onboard from hydrogen via a fuel cell, powering the electric motor.

EVs are significantly more energy-efficient than ICE cars, both now and in future projections (see Figure 1). ICE vehicles lose most energy as heat and often operate inefficiently. BEVs are the most efficient, followed by hydrogen FCEVs. Overall, BEVs can be up to four times more efficient than ICE vehicles.

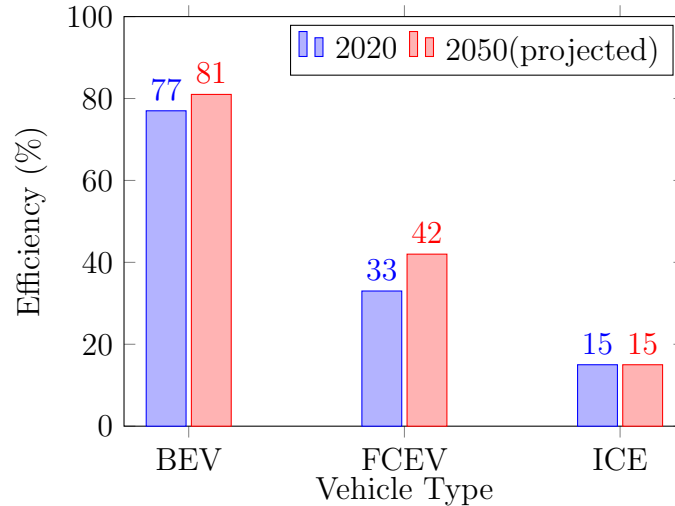


Figure 1: Comparison of well-to-wheel efficiencies for different vehicle types (2020 vs. projected 2050).

**Buses and trains:** Trains are mostly electrified today and buses are increasingly moving towards electrification.

- **Trains:** electric trains have been widely adopted in Europe; less so in New Zealand. Electrification is straightforward where overhead lines exist. In non-electrified routes, there are some alternatives including hydrogen trains (e.g. Augsburg–Füssen route in Germany, in service since 2024 and Königstein and Frankfurt since 2022).
- **Buses:** Electrification of city buses is accelerating rapidly. The EU aims for 100% electric bus sales by 2030, a target that aligns with New Zealand’s goals: aiming to only purchase zero-emission buses from 2025 and achieve full decarbonisation by 2035.

**Trucks:**

- **Weight constraint myth:** Battery-electric trucks, even in the 40-tonne class with 800 km range, are becoming electrified.
- **Cost comparison:** Total cost of ownership (TCO) studies show that electric trucks are cost-competitive with diesel.

**Aviation:** Depending on distance, sustainable aviation can look different.

- Short-haul flights may be replaced by electric planes or alternative transport like high-speed rail or cars.
- Mid-haul: Partial electrification is plausible.
- Long-haul: Options include sustainable aviation fuels (SAF) or hydrogen. SAF can be produced from biomass (Bio-SAF) or green hydrogen + captured CO<sub>2</sub> (e-SAF). In hydrogen-powered aircrafts, there can be combustion or fuel cell–electric configurations, however volume limitations remain a major design challenge.

**Broader context and NZ perspective:** There are some ongoing debates in the world, particularly, in Europe:

- **Trains vs short flights:** Banning short-haul flights under 2.5 hours, encouraging railway alternatives and limiting public sector air travel. For example, initiatives like Germany’s €49 monthly transport pass aim to boost public transport.
- **ICE car phase-outs:** EU target: Ban ICE sales by 2035. Some countries oppose outright bans and advocate for technological neutrality (“let the market decide”).

In the case of New Zealand, the Clean Car Rebate (which incentivised low-emission vehicles and penalised high-emission ones) has been discontinued, affecting affordability/competitiveness of EVs.